

Introduction to organic chemistry

Risk and Hazard

Term	Definition	Example
Hazard	Something with the potential to cause harm.	Concentrated hydrochloric acid is corrosive .
Risk	The likelihood (chance) that a hazard will actually cause harm.	The risk is low if you are wearing gloves and safety glasses, but high if you are handling it with bare hands.

Why are risk assessments necessary?

Many organic compounds are hazardous (e.g. flammable, toxic, corrosive). A risk assessment is a proactive process to ensure safety before starting any practical work.

Definition of Risk Assessment:

A risk assessment is the identification of the hazards involved in a procedure and the implementation of control measures needed to reduce the risks from those hazards.

The Process:

Identify the Hazards: What chemicals are you using? (e.g. Benzene is carcinogenic, ethanol is flammable).

Decide who might be harmed and how: (e.g. the experimenter, people nearby).

Evaluate the risks and implement control measures: What can you do to make the procedure safer?

Ways to Reduce Risks & Carry Out Reactions Safely

i) Working on a smaller scale

—What it means: Using the smallest possible quantities of chemicals to achieve the desired reaction.

—Why it reduces risk: Less chemical = less energy released if something goes wrong = smaller spill = reduced exposure = lower risk.

—Example: Using 0.1 mol instead of 10 mol of a reactant in a synthesis.

ii) Taking precautions specific to the hazard

—What it means: Using control measures that directly address the identified hazard.

—Why it reduces risk: It targets the source of the danger directly.

—Examples:

Flammable Hazard: Use a water bath for heating instead of a Bunsen burner; avoiding naked flames in the lab; heat in an inert atmosphere.

Toxic / Corrosive Hazard: Use a fume cupboard to prevent inhalation of vapours. Wear appropriate Personal Protective Equipment (PPE) like gloves, lab coat, oxygen mask, and safety goggles.

Corrosive Liquid Spill: Have a spill kit nearby (e.g. neutralising agent for acids).

iii) Using an alternative method that involves less hazardous substances

—What it means: Finding a different chemical reaction or pathway that achieves the same product but uses safer chemicals.

—Why it reduces risk: Eliminates or reduces the hazard at the source.

—Examples:

Using ethanol as a solvent instead of more toxic methanol.

Using a green catalyst like an enzyme instead of a strong acid

Lab Equipment Hazards & Alternatives

Apparatus	Main Hazards	Safer Alternatives
Mercury Thermometer	• Toxic if broken • Hard to clean	• Digital Thermometer • Alcohol Thermometer
Bunsen Burner	• Open flame (fire risk) • Gas leaks • Causes burns	• Electric Heaters • Water Bath / Oil Bath

Common hazard symbols

Moderate health hazard	Serious health hazard	Toxic (poisonous/ lethal)
Causes skin irritation	Causes breathing difficulties/ carcinogenic	Could cause death if swallowed or inhaled
Flammable	Oxidising	Harmful to the environment
Corrosive	Explosive	

—**Homologous series:** a family of compounds with similar chemical properties with the same functional group, same general formula, which differs in molecular formula by CH_2 from the next member, and shows a gradual change in physical properties.

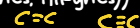
—**Functional group:** an atom or group of atoms in a structure (molecule/compound) that is responsible for the chemical reactions of a compound.

NAME	GENERAL FORMULA	EXAMPLE
alkane	$\text{C}_n\text{H}_{2n+2}$	CH_4
alkene	C_nH_{2n}	C_2H_4
halogenoalkane	$\text{C}_n\text{H}_{2n+1}\text{X}$	$\text{CH}_3\text{CH}_2\text{Br}$
alcohol	$\text{C}_n\text{H}_{2n+1}\text{OH}$	$\text{CH}_3\text{CH}_2\text{OH}$

Hydrocarbons

Hydrocarbon: organic compounds that contain only carbon and hydrogen.

Unsaturated: contains at least one carbon-carbon double bond (less than the maximum amount of hydrogen (e.g. Alkenes, Alkynes))



Saturated: contain the all carbon-carbon bonds are single (maximum amount of hydrogen (e.g. Alkanes))



Types of formulae

1. Molecular Formula

Shows the actual number of each type of atom in one molecule.

Does not show connections or structure.

Example: C_6H_{14} (hexane)

2. Empirical Formula

Simplest whole-number ratio

Example: C_3H_7 (from C_6H_{14})

3. Structural Formula

Shows connections briefly where single bonds are not shown

Example: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

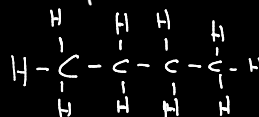
Note: branches that containing more than one atoms are put in brackets

Example: $\text{CH}_3\text{CH}_2(\text{CH}_3)\text{CH}_2\text{CH}_3$ (2-methylbutane)

4. Displayed:

Shows all atoms and bonds

Example:



5. Skeletal Formula

Simplified representation showing carbon skeleton

Characteristics:

Carbon atoms not shown — represented by line ends/vertices

Hydrogen atoms on carbon not shown

Heteroatoms (O, N, Cl, etc.) ARE shown

Each line represents a C—C bond

Rules:

Carbon atoms at every:

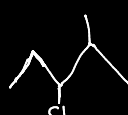
Line end

Vertex/corner

Junction point

Double/triple bonds shown as double/triple lines

Example:



3-Chloro-2-methylpentane

6. 3D Formula

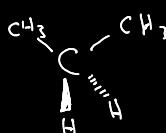
Shows spatial arrangement

Wedge = toward you

Dashed = away from you

Line = in plane

Example:



Practice Drawing These Compounds in All Formula Types:

1. Butane 2. Propanoic Acid 3. Cyclohexane

4. 2-methylpropane 5. Ethanol 6. Ethene

7. Propanone 8. 2-chlorobutane

Introduction to organic chemistry

Naming rules

IUPAC System: International standard for naming compounds

Two Main Parts:

-1. Parent Chain Identification

Rules:

Longest continuous carbon chain

Determines base name

Parent Chain Reference Table:

Carbons	Prefix	Parent Name
1	meth-	methane
2	eth-	ethane
3	prop-	propane
4	but-	butane
5	pent-	pentane
6	hex-	hexane
7	hept-	heptane
8	oct-	octane
9	non-	nonane
10	dec-	decane

-2. Substituent Naming

A. Alkyl Groups

Carbon chains NOT part of parent

Naming: Replace "-ane" with "-yl"

Common Alkyl Groups:

Alkane	Alkyl Group	Structure
Methane	Methyl	$-\text{CH}_3$
Ethane	Ethyl	$-\text{CH}_2\text{CH}_3$
Propane	Propyl	$-\text{CH}_2\text{CH}_2\text{CH}_3$
Butane	Butyl	$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

B. Halogens (Halogenoalkanes)

Always named as PREFIXES

Halogen: Prefixes

F:	fluoro-
Cl:	chloro-
Br:	bromo-
I:	iodo-

Naming

Step 1: Identify Parent Chain

Find **LONGEST** continuous carbon chain

This determines base name (hexane, heptane, etc.)

Step 2: Number the Parent Chain

Number from end that gives **SUBSTITUENTS** lowest numbers

Compare both directions, choose lowest number set

Step 3: Identify & Name Substituents

Alkyl groups: methyl, ethyl, propyl, etc.

Halogens: fluoro, chloro, bromo, iodo

Numbering Rules

Primary Rule: Number parent chain to give substituents **LOWEST** possible numbers (lowestsum)

Compare numbering from both ends

Choose direction giving smallest numbers

Special Cases:

In case of tie, prioritize **alphabetical** order

Name Assembly Formula

Format: (Number)-(Substituent)(Parent Name)

Punctuation Rules:

Numbers separated by commas

Numbers and words separated by hyphens

No spaces in final name

Multiples of Same Substituent:

Use multiplying prefixes:

di- (2)

tri- (3)

tetra- (4)

penta- (5)

Alphabetical Order:

List substituents alphabetically

Ignore multiplying prefixes (di-, tri-) when alphabetizing

Naming Alkenes, Alcohols and Alkynes

Step 1: Identify the Parent Chain

Must be the longest continuous carbon chain that **CONTAINS** the functional group (double bond or The carbon atom bonded to the -OH group)

Not necessarily the longest overall chain if that chain doesn't include the functional group.

Step 2: Number the Parent Chain

Priority Rule: Number the chain to give the functional group (-OH or C=C) the lowest possible number. This takes priority over substituent positions.

Alcohols: The position of the -OH group is indicated by this number.

Alkenes: Indicate the bond's position with the lower-numbered carbon atom involved in the double bond.

Step 3: Name Assembly

Format: (Substituent Positions)-(Substituent Names)-(Parent Name with Functional Group Position and Suffix)

Alcohols:

Suffix: End of the parent alkane name with "-ol".

Example 1: $\text{CH}_3\text{CH}_2\text{OH}$ → Parent chain: ethane → ethan-1-ol, which simplifies to ethanol (as it is the only isomer)

Example 2: $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ → Parent chain: propane. Numbering gives the -OH the lowest number (carbon 2). Propan-2-ol.

Example 3: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ → Propan-1-ol.

Alkenes:

Suffix: Replace the final "-ne" of the parent alkane "-ene" for alkenes (C=C)

Example 1: $\text{CH}_2=\text{CH}_2$ → Parent: ethane → ethene.

Example 2: $\text{CH}_3\text{CH}=\text{CH}_2$ → Parent: propane. Double bond starts at carbon 1 → Prop-1-ene.

Example 3: $\text{CH}_3\text{CH}=\text{CHCH}_3$ → Parent: butane. Double bond starts at carbon 2 → But-2-ene.

Important Note: No Adjacent Vowels

When adding a suffix, drop the final vowel of the parent name if the suffix starts with a vowel.

Single functional group (suffix starts with vowel):

butane + -ene → butene (not butaene) ("-ne" is always removed when adding "ene")

propane + -ol → propanol (not propaneol)

Multiple functional groups (prefix like di-, tri- added):

The suffix now starts with a consonant (d, t), so keep the parent's vowel.

pentane + diene → penta~~a~~diene ("-ne" is always removed when adding "ene")

ethane + diol → ethanediol

Simple Rule: Parent vowel + suffix vowel **✗** → drop parent vowel.

Parent vowel + consonant **✓** → keep parent vowel.

Functional Group Priority for Naming

When a molecule has multiple functional groups, the one with the highest priority determines the suffix.

Functional Group	Class Name	Suffix (in Highest Priority)	Prefix (in Lower Priority)
-COOH	Carboxylic Acid	-oic acid	---
-OH	Alcohol	-ol	hydroxy-
C=C	Alkene	-ene	en-
-X (F, Cl, Br, I)	Haloalkane	---	fluoro-, chloro-, etc.
-CH ₃ , -C ₂ H ₅	Alkane/Alkyl	-ane	alkyl-

IUPAC Nomenclature: Cycloalkanes

1. Basic Cycloalkane Naming

Rule: Add prefix "cyclo-" to parent alkane name

Carbon Atoms	Name	Structure
3	cyclopropane	
4	cyclobutane	
5	cyclopentane	
6	cyclohexane	
7	cycloheptane	
8	cyclooctane	

2. Numbering Cycloalkanes with Substituents

Rule 1: Number to give substituents the **lowest possible numbers**

Rule 2: If only one substituent, no number needed

Rule 3: If multiple substituents, number to achieve **lowest set of numbers (sum)**

Rule 4: List substituents in **alphabetical order**

Note: The general formula for cycloalkanes is C_nH_{2n} .

Why? Start from an alkane formula (C_nH_{2n+2}).

When you form a **ring** in the carbon chain, you lose 2 hydrogen atoms, giving C_nH_{2n} .

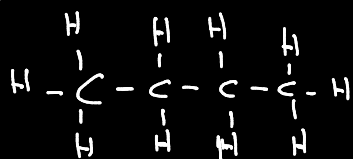
Extended rule:

Each ring $\rightarrow$ subtract 2 H.

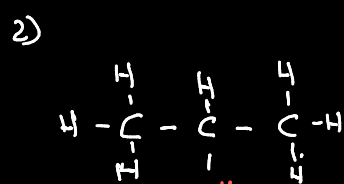
Each double bond $\rightarrow$ subtract 2 H.

Each triple bond $\rightarrow$ subtract 4 H (because it acts like two double bonds in hydrogen count).

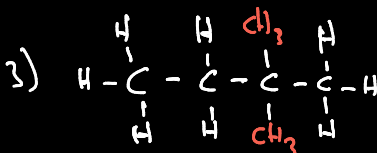
Examples



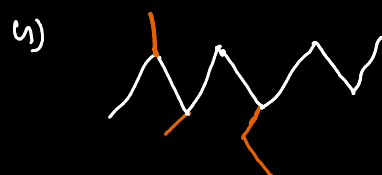
Butane



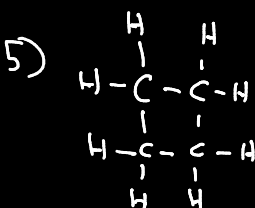
2-methylbutane



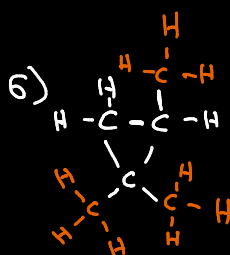
2,2-dimethylbutane



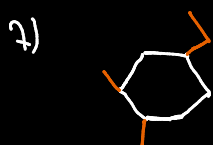
5-ethyl-2,3-dimethyloctane



Cyclobutane

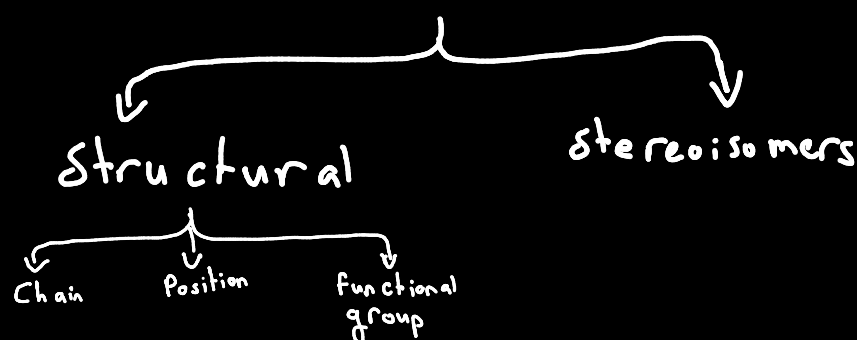


1,1,2-dimethylcyclopropane



1-ethyl-3,4-dimethylcyclohexane

Types of isomers



Structural isomers: Compounds with the same molecular formula but different structural arrangements of their atoms.

-Chain Isomers

Difference: The carbon skeleton (the chain) is **arranged differently**

Cause: Branching of the carbon chain.

Example: C_4H_{10}

Butane: $CH_3-CH_2-CH_2-CH_3$ (straight chain)

2-Methylpropane: $(CH_3)_2CH-CH_3$ (branched chain)

-Position Isomers

Difference: The position of a **functional group** or substituent on the carbon chain changes.

Cause: The functional group is attached to a different carbon atom.

Example: C_3H_7Cl

1-Chloropropane: $CH_3-CH_2-CH_2-Cl$

2-Chloropropane: $CH_3-CH(Cl)-CH_3$

-Functional Group Isomers

Difference: The molecules contain different functional groups.

Cause: The atoms are connected in ways that form different functional groups.

Example: C_2H_6O

Ethanol: CH_3-CH_2-OH (alcohol)

Methoxymethane: CH_3-O-CH_3 (ether)

Common Functional Group Isomer Pairs:

Alkene & Cycloalkane: C_3H_6 (Propene & Cyclopropane)

Alcohol & Ether: C_3H_8O (Propanol & Methoxyethane)

Aldehyde & Ketone: C_3H_6O (Propanal & Propanone)

Carboxylic Acid & Ester: $C_2H_4O_2$ (Ethanoic acid & Methyl methanoate)

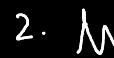
note: for functional groups to be isomer pairs they must have the same general formula

E.g.

- C_6H_{14}



hexane



2-methylpentane



3-methylpentane



2,3-dimethylbutane



2,2-dimethylbutane

Alkanes

Alkanes are **saturated hydrocarbons** with only single bonds hence containing the maximum amount of hydrogen.

General formula: C_nH_{2n+2}

Properties:

- 1—**Insoluble** in water due to their non-polar nature, but soluble in non-polar organic solvents
- 2—**Flammable** (Larger alkanes are less volatile and burn less easily, making them less flammable than smaller alkanes)
- 3—**Unreactive** (e.g. oxidation) due to being non polar and strong C—C and C—H sigma bonds
- 4—*States (C_1 to C_4 : Gases / C_5 to C_{17} : Liquids / C_{18} and above: Waxy Solids)
- 5—*Boiling/Melting Points increases with chain length due to having more electrons(stronger IMF)
- 6—They become thicker/more Viscous as chain size grows
- 7—**Colourless & odourless**
- 8—**Non-toxic**

Extraction

1—**Fractional distillation**

Objective: To **separate complex crude oil** (a mixture of hydrocarbons) into simpler, more useful fractions based on differences in their boiling points.

Basic Principle:

Hydrocarbons have different sizes:

Small molecules (e.g. propane, butane) have weak intermolecular forces, resulting in low boiling points.

Large molecules (e.g. those in tar) have strong intermolecular forces, resulting in high boiling points.

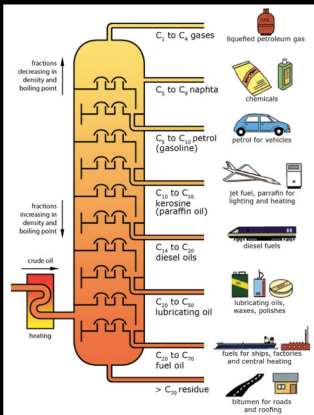
This difference allows them to be separated by heating

Why is it Important?

- a)Converts a useless raw material (crude oil) into valuable products.
- b)Provides the fuels that power modern transport (cars, planes, trucks).
- c)Provides the feedstock (raw materials) for the petrochemical industry to make plastics, solvents, and detergents.

Description:

- Heating** Crude oil is first heated in a furnace, turning it into a **hot vapor** and liquid mixture.
- Entering the Column:** The hot mixture is pumped into the bottom of a **fractionating column** (a tall tower containing trays or plates).
- ***Separation by Temperature Gradient:**
 - The column is hot at the bottom and cool at the top.
 - As the vapors rise, they cool down.
 - Each hydrocarbon vapour **condenses** (turns back into liquid) **at a different level**(temp) in the column, depending on its specific boiling point.
 - **The higher the boiling temperature the lower the level at which it condenses**
- Collection:** The trays in the column collect the condensed liquids. Each tray contains a different fraction, which is then piped out for further processing and use.



2—Cracking

Objective: To break down large, **less useful heavy** hydrocarbons from crude oil (like gas oil or lubricating oil) into **smaller, more valuable** hydrocarbons (like petrol and alkenes).

Basic Principle:

There is a high market demand for shorter-chain products (like gasoline for cars). However, fractional distillation naturally produces a surplus of longer-chain products.

Problem: Supply of large molecules > Demand for them.

Solution: Cracking — We break, or "crack," the large molecules into smaller, more in-demand ones.

Why is Cracking So Important?

- a)Meets Demand for Petrol: Increases the yield of gasoline (petrol) from a barrel of crude oil.
- b)Produces Alkenes: Creates alkenes (e.g. ethene, propene), which are essential feedstock for the petrochemical industry to make plastics (like polyethene), solvents, and many other chemicals.
- c)Balances Supply & Demand: Converts less useful heavy fractions into more profitable, high-demand products.

General Word Equation:

Long-Chain Hydrocarbon → Smaller Alkane + Alkene

Catalytic Cracking

Conditions: **high temperature** (around 500°C) and slight pressure, in the presence of **aluminium oxide** or zeolite catalyst (a porous aluminosilicate).

Why it's Preferred:

- Faster reaction rate.
 - Lower energy costs (lower temperature compared to thermal cracking).
 - Produces a higher proportion of branched-chain hydrocarbons, which improve the octane rating of petrol (making it burn more smoothly in engines).
- Products: High yield of petrol (gasoline) and some alkene

Example Equations

Worked example:

$C_9H_{20} \rightarrow C_7H_{16} + ?$

Step 1: Balance Carbon Atoms.

Left side: 9 Carbon atoms.

Right side so far: 7 Carbon atoms (from heptane).

Missing product must have: $9 - 7 = 2$ Carbon atoms.

Step 2: Balance Hydrogen Atoms.

Left side: 20 Hydrogen atoms.

Right side so far: 16 Hydrogen atoms (from heptane).

Missing product must have: $20 - 16 = 4$ Hydrogen atoms.

Step 3: Identify the Missing Product.

We have a molecule with 2 Carbon atoms and 4 Hydrogen atoms. This is C_2H_4 .

Check its type: For 2 carbons, an alkane would be C_2H_6 . C_2H_4 follows the formula C_nH_{2n} , so it is an alkene.

The Final Answer:

$C_9H_{20} \rightarrow C_7H_{16} + C_2H_4$

Nonane → Heptane + Ethene

—Example 1: Cracking Decane ($C_{10}H_{22}$)

$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$

(Decane → Octane + Ethene)

—Example 2: Cracking Dodecane

$C_{12}H_{26} \rightarrow C_{10}H_{22} + C_2H_4$

Dodecane → Decane + Ethene

3—Reforming

Objective: To convert straight-chain alkanes into branched-chain alkanes and cyclic or aromatic hydrocarbons (like benzene) to improve the quality of petrol (gasoline).

Basic Principle:

Problem: Straight-chain hydrocarbons in the gasoline fraction (e.g. heptane, C_7H_{16}) burn poorly in car engines, **causing "knocking"** (uncontrolled explosion).

Solution: **Reforming** rearranges the molecular structure without breaking the molecules into smaller pieces. This process changes the structure but not the size of the molecules.

Key Change: Straight-Chain Alkane → Branched-Chain Alkane / Cycloalkane

Why is Reforming So Important?

—**Increases Octane Number:** Branched-chain and aromatic hydrocarbons burn much **more smoothly, more efficiently and prevent knocking**, which is measured by a higher octane rating. High-octane fuel is essential for modern high-performance car engines.

Example 1:

Hexane → Benzene + Hydrogen

$C_6H_{14} \rightarrow C_6H_6 + 4H_2$

Alkanes

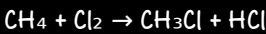
Alkane reactions

Free Radical Substitution with Halogens(Chlorine)

It could also called halogenation(e.g. chlorination)

Reaction: Alkane + Chlorine → Chloroalkane + Hydrogen Chloride
(in the presence of UV light)

Example (with Methane):



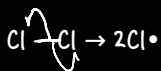
Alkanes are **nonpolar and have strong C—C and C—H bonds**, making them unreactive towards ions. Chlorine molecules (Cl₂) are also nonpolar. This reaction proceeds via a **free radical mechanism**—a species with an unpaired electron—which is **highly reactive**.

The Mechanism: A Chain Reaction

The mechanism occurs in three distinct stages: **Initiation**, **Propagation**, and **Termination**.

Stage 1: Initiation (The Start)

UV light provides energy to break the (relatively weak) Cl—Cl bonds. This splitting is called **homolytic fission**, where each atom gets one electron from the bond, forming two highly reactive **chlorine free radicals**.

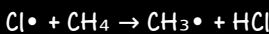


Free radical: A **species** (atom or a group of atoms) which have an **unpaired electron** which results from the **homolytic fission** where electrons are **shared equally**/evenly between the two atoms.(making them highly reactive)

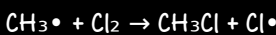
Stage 2: Propagation (The Chain—Carrying Steps)

The chlorine **radicals** produced are very reactive. They attack the alkane molecules in a cycle that **generates new radicals**.

—Step 1: A chlorine radical attacks a methane molecule, pulling off a hydrogen atom. This forms hydrogen chloride (HCl) and leaves behind a methyl radical (•CH₃).



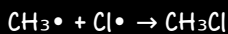
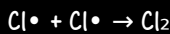
—Step 2: The methyl radical then collides with a chlorine molecule (Cl₂). It takes a chlorine atom to form the product, chloromethane (CH₃Cl), and **regenerates a new chlorine radical**.



This new chlorine radical can then go back to Step 1 and repeat the process, creating a self—sustaining chain reaction.

Stage 3: Termination (The Stop)

The chain reaction stops when **two radicals** combine, forming a stable molecule. This consumes the radicals without generating new ones.



Further (Multiple) Substitution

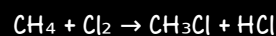
Further substitution occurs when more than one hydrogen atom has been replaced by a chlorine atom.

This is a major **limitation** of the reaction, making it difficult to obtain a single, pure product.

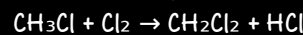
Why Does it Happen?

The primary product (e.g. CH₃Cl) is still an alkane and has C—H bonds that can be attacked by these radicals. The C—H bonds in the product are actually slightly weaker than in the original alkane, making further substitution statistically likely. The Reaction Sequence with Methane

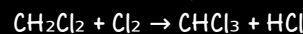
When methane (CH₄) reacts with chlorine, a mixture of four products is formed.



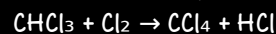
(Chloromethane)



(Dichloromethane)

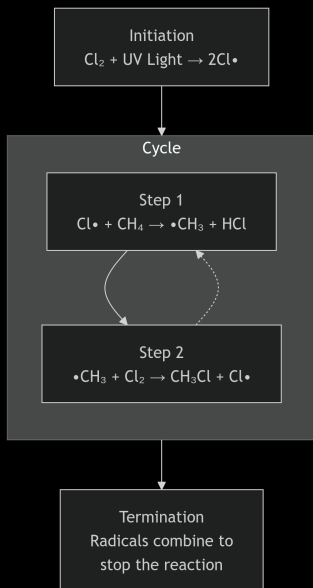


(Trichloromethane)



(Tetrachloromethane)

Type & Mechanism:
Free redial Substitution



The Overall Problem: A Mixture of Products

You do not get just CH₃Cl. The final product mixture contains:

Unreacted CH₄

CH₃Cl (Desired product)

CH₂Cl₂

CHCl₃

CCl₄

HCl

Combustion

Combustion Reactions of Alkanes

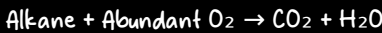
Combustion is the reaction of a fuel with oxygen. There are two types: complete and incomplete combustion.

a) Complete Combustion

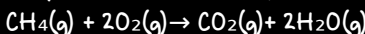
—Conditions: **excess oxygen**.

—Products: Carbon dioxide (CO₂) and Water (H₂O).

—Example (for any alkane):



—Example (for Methane, CH₄):



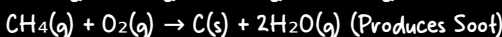
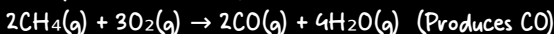
b) Incomplete Combustion. (e.g. in an enclosed space)

—Conditions: **Limited oxygen**.

—Products: Carbon monoxide (CO) and/or Carbon (soot, particulates), and Water (H₂O).

Unburned hydrocarbons may also be released.

Examples (for Methane):



Pollutants from Alkane Combustion

Pollutant	How it is Formed	Problem it Causes
Carbon Monoxide (CO)	Incomplete combustion due to insufficient oxygen.	Toxic/Poisonous: It binds irreversibly to haemoglobin in red blood cells, preventing them from carrying oxygen. This leads to oxygen deprivation in the body, causing fainting, coma, or even death.
Oxides of Nitrogen (NO _x) e.g., NO, NO ₂	Formed when nitrogen (N₂) and oxygen (O₂) from the air react at the high temperatures in an engine.	Acidity & Toxicity: - They dissolve in water in the atmosphere to form nitric acid (HNO₃), a component of acid rain. - Acid rain damages buildings, statues, and harms aquatic life and forests. - They also cause respiratory problems in humans.
Oxides of Sulfur (SO ₂)	Formed from the combustion of sulfur impurities present in fossil fuels.	Acidity: - It dissolves in water to form sulfurous acid (H₂SO₃), which oxidizes to sulfuric acid (H₂SO₄). This is a major component of acid rain. - Acid rain damages buildings, statues, and harms aquatic life and forests.
Carbon Particulates (Soot)	Incomplete combustion , forming solid carbon particles.	- Causes respiratory problems (asthma, lung damage). - Contributes to smog, reducing visibility.
Unburned Hydrocarbons	Released due to incomplete combustion .	- Some are toxic or carcinogenic. - Contributes to the formation of ground-level ozone (smog), which causes breathing difficulties.

Catalytic converter

Purpose

Catalytic converters are fitted to vehicle exhausts to reduce the emission of harmful pollutants produced by the combustion of alkane fuels in the engine.

How They Work

They contain a honeycomb structure made of a ceramic or metal, providing a large surface area.

This structure is coated with a thin layer of precious metal catalysts such as **Platinum (Pt)**,

Palladium (Pd), and **Rhodium (Rh)**.

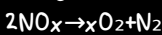
As the hot exhaust gases pass over the catalyst, redox (reduction—oxidation) reactions occur on the surface, converting the toxic pollutants into less harmful gases.

Key Chemical Reactions

The catalytic converter facilitates two main types of reactions:

a) Reduction of Nitrogen Oxides (NO_x)

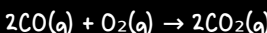
Nitrogen oxides are reduced to harmless nitrogen gas.



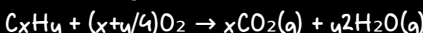
A more specific example: $2\text{NO}(\text{g}) + 2\text{CO}(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{CO}_2(\text{g})$

b) Oxidation of Carbon Monoxide and Unburned Hydrocarbons

Carbon monoxide is oxidized to carbon dioxide.



Unburned hydrocarbons are oxidized to carbon dioxide and water.



Reducing Sulfur Oxides (SO_x)

1.Desulfurization of fuels before combustion. (reaction with CaO)

2.Using low—sulfur and ultra—low—sulfur fuels.

Calcium hydroxide is used to neutralise HCl / SO₂ / NO_x / CO₂ (as it is basic)

Pollution and Alternatives

The Greenhouse Effect

1. The Natural Greenhouse Effect

The Greenhouse Effect is a natural process that is essential for life on Earth.

How it works:

Short-wave radiation (visible light, UV) from the sun passes through the Earth's atmosphere and warms the Earth's surface.

The Earth radiates this energy back as long-wave radiation (infrared, heat).

Greenhouse Gases (GHGs) in the atmosphere, such as Carbon Dioxide (CO₂), Methane (CH₄), and Water Vapour (H₂O), **trap** some of this outgoing heat.

This trapped heat warms the planet, keeping it at a stable and habitable average temperature.

Without the natural greenhouse effect, Earth would be too cold for life.

2. Enhanced Greenhouse Effect & Global Warming

The problem is the enhanced greenhouse effect caused by human activities.

Global Warming refers to the long-term rise in the Earth's average surface temperature due to this enhanced effect.

How it happens:

Human activities have significantly **increased the concentration of greenhouse gases** in the atmosphere.

A thicker layer of GHGs **traps more and more of the outgoing infrared radiation**.

This excess trapped heat causes the Earth's **average temperature to rise**, leading to climate change.

3. Main Causes (Human Activities)

Burning of Fossil Fuels:

Coal, oil, and natural gas are burned for electricity, heat, and transport.

This **releases vast amounts of CO₂** that had been stored underground for millions of years.

Example: Combustion of methane: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

Deforestation:

Trees absorb CO₂ from the atmosphere for photosynthesis. Cutting them down removes these "carbon sinks."

The ability for the biosphere to **absorb CO₂ is reduced**

Burning forests to clear land releases the stored carbon directly as CO₂.

Agriculture:

Livestock (e.g. cattle) produce **large amounts of Methane (CH₄)** during digestion.

The use of nitrogen-based fertilizers releases Nitrous Oxide (N₂O), a powerful greenhouse gas.

4. Effects

Rising Sea Levels — from **melting ice** and warmer water expanding. Floods coastal areas.

Extreme Weather — **more** heatwaves, droughts, floods, and stronger storms.

Loss of Habitats — ice-dependent species (like polar bears) **lose their homes**. Coral reefs die from warm water (coral bleaching).

Ocean Acidification — oceans absorb more CO₂, becoming more acidic. This harms shellfish and coral.

Electric Cars/devices

Advantages

They do not emit CO₂ which causes global warming

it is powered by electricity which is a renewable resource

do not emit substances that would damage and pollute such as nitrogen oxides

They are much quieter and reduce noise pollution

Disadvantages

Electricity often comes from fossil fuels

Limited range compared to gasoline

Charging takes longer than gassing up

Alternative fuels

1. Reasons for Developing Alternative Fuels

The shift to alternative fuels is driven by two main factors: sustainability and reducing emissions.

a) Reducing Emissions

Problem: Burning fossil fuels (like petrol) contribute to global warming releases large amounts of pollutants.

Carbon Dioxide (CO₂): A major greenhouse gas, leading to climate change.

Other Pollutants: Sulfur dioxide (SO₂) causes acid rain, and nitrogen oxides (NO_x) contribute to smog.

Solution: Alternative fuels that are **carbon neutral or close to being carbon neutral**.

b) Sustainability

Problem: Fossil fuels are **non-renewable**. They will eventually run out.

Solution: Alternative fuels are often based on renewable resources (e.g. plants, water), making our energy supply more secure.

2. Carbon Neutrality

Definition: A process is carbon neutral if the **net release of carbon dioxide (CO₂)** into the atmosphere over its entire lifecycle is **zero**.

—The Core Idea: The amount of **CO₂ absorbed** from the atmosphere during the production of the fuel(photosynthesis) is **equal** to the amount of **CO₂ released** when the fuel is used.

Example: A Tree

CO₂ Absorbed: A tree absorbs CO₂ as it grows.

CO₂ Released: If the tree is burned, it releases the same amount of CO₂ it absorbed.

Net Change: Zero. This is carbon neutral.

The Key Limitation: This concept often ignores other greenhouse gases (like methane from farming) and the energy used for processing and transport, which usually involves fossil fuels.

Biofuels

—Description: **Fuels derived** directly from **living** or recently living matter (biomass), such as plants (e.g. corn, sugarcane, soy) or organic waste.

—Key Types:

Bioethanol: An alcohol made **by fermenting** plant sugars(using enzymes or bacteria), often blended with petrol.

Biodiesel: Made from vegetable oils or animal fats, used as a direct substitute for diesel.

—Core Principle: They are considered **carbon neutral** because the CO₂ released during combustion was absorbed from the atmosphere by the plants as they grew.

Hydrogen

—Description: Hydrogen gas (H₂) is used as an energy carrier, either by **burning** it in an engine or, more efficiently, in a **fuel cell**.

—How a Fuel Cell Works: It **combines hydrogen and oxygen to produce electricity**, with water (H₂O) as the only by-product.

—Core Principle: It is a **zero-carbon fuel** at the point of use. However, its environmental benefit depends entirely on how the hydrogen is produced.

Fuel Type	Description	Advantages	Disadvantages
Petrol / Diesel	Conventional fuels from crude oil.	- High energy density. - Well-established infrastructure & engines. - Reliable and cheap to produce.	Not renewable. - High CO₂ & pollutant emissions (CO, NO_x, SO₂). - Contributes significantly to climate change & air pollution.
Natural Gas (CNG/LNG)	Compressed or Liquefied Methane from fossil reserves.	Cleaner burning than petrol (fewer particulates and SO₂). - Often cheaper than petrol.	Still a fossil fuel — produces CO₂. - Methane leaks are a potent greenhouse gas. - Requires high-pressure storage tanks.
Biofuels (e.g. Bioethanol, Biodiesel)	Fuels derived from plant matter or animal waste.	Renewable resource. - Theoretically carbon neutral. - Reduces reliance on crude oil.	Uses a lot of land (leads to deforestation & habitat loss). Food vs. fuel conflict (uses crops/land that could grow food). Limited yield per hectare of land. Uses energy for manufacturing and transport (making it not fully carbon neutral).
Hydrogen	Can be burned directly or used in a fuel cell to produce electricity.	Zero CO₂ emissions at point of use (only produces water). - Very high efficiency in fuel cells. - Can be produced from water (electrolysis).	Production is rarely green (often from fossil fuels). - Difficult and expensive to store and transport. - Highly flammable. Limited infrastructure.

Clarification:

Although hydrogen fuel does not produce carbon dioxide (CO₂) when used, it does release water vapor (H₂O), which is a greenhouse gas.